
Bearing Health Monitor

Retrofit wireless vibration monitoring that names **which bearing element is failing** — physics-first, no training data. Built and supported in Kazakhstan.

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Pre-seed / working prototype · pre-revenue

seeking a commercial co-founder

Bearings fail slowly, then all at once

And most plants find out at the second stage

Rolling-element bearings degrade **progressively** — microscopic spalling → characteristic defect frequencies → heat → seizure.

The failure is detectable in advance from vibration.

Yet mid-tier and legacy plants still run **reactive or fixed-interval** maintenance: replace on a calendar, or replace after it breaks.

A single unplanned stop on a critical line — a mill stand, a main pump, a furnace blower — costs **tens of thousands of dollars per hour**, plus collateral damage and safety exposure.

Why the gap persists

- Premium monitoring platforms are priced and contracted for tier-1 plants.
- Route-based manual measurement is quarterly at best — faults develop between rounds.
- Systems that need historical training data are useless on day one.
- Foreign vendors sell in English, invoice in EUR, and support from another time zone.

Physics, not a black box

Explainable on day one, with zero training data

Bearing geometry and shaft speed determine four characteristic defect frequencies exactly (BPFO / BPFI / BSF / FTF). A defect on a given element rings at *its* frequency and harmonics. Find the comb, name the element.



Works from day one

No historical data, no model training, no per-machine tuning. Deployable on a plant that has never measured anything.

Every verdict is auditable

"Outer race, 12x over your own baseline, sustained across 3 captures."
A plant engineer can check the reasoning.

ML is upside, not a prerequisite

Anomaly and remaining-useful-life models layer on later as field data accumulates — the data moat.

What already exists

Public repo, 40+ passing tests, live in-browser demo

COMPONENT	STATE
Defect-frequency + physics-line model $2 \times \text{BSF} \pm \text{FTF}$, $\text{BPFI} \pm \text{fr}$	done
Envelope demodulation + comb SNR	done
Matched kurtogram band selection	done
Bearing-slip alignment	done
Baseline trending + sustained alarm	done
Live browser demo (synthetic machine)	done
Trilingual site EN / RU / KK	done
Field rig (Teensy + ADXL1002)	parts in hand
Wireless node, gateway, cloud	not started

Unusual for this stage

The detection core is open and reproducible. Anyone can clone the repo, download the public datasets and re-run every number in this deck — `pytest tests/`.

The browser demo runs the *same* algorithm as the product, compiled to JavaScript. No video, no mockup — a reviewer can drive it.

github.com/alibek-git/bearing-health-monitor
bearing-monitor.pages.dev

Measured results

CWRU seeded faults and NASA IMS run-to-failure, shipped default configuration

CWRU — 1772 RPM DRIVE END

RECORD	TRUTH	VERDICT	COMB SNR
98	normal	healthy	3.95
106	inner race	faulted — BPFI	35
131	outer race	faulted — BPFO	141
136	outer race	faulted — BPFO	39
119	ball	suspect — element wrong	6.1

NASA IMS SET 2 — RUN TO FAILURE, 7 DAYS

BEARING	SUSTAINED ALARM	PEAK SNR
1 — fails (outer race)	idx 59 / 99	31.6
2 — survives	idx 90	33.2
3 — survives	idx 91	25.2
4 — survives	idx 88	38.5

The failing bearing raises a sustained alarm **~2.5 days** before it dies, and **~2 days** before the earliest neighbour. All four share a shaft and a BPFO, so the neighbours alarm late from casing crosstalk — the claim is **lead time**, not neighbour silence.

What this evidence does *not* support

Stated here because you would find it in the repo anyway

No false-alarm rate

One healthy record, reading 3.95 against a warn threshold of 4.0 — a 1.2% margin. A rate needs a healthy population.

No lead-time distribution

$n = 1$ failure event. One observation has no dispersion, so 2.5 days has no lower bound — and nothing supports the word *weeks*.

Ball faults unsolved

Detected as a level change, but the element ranking is wrong. Race faults — the majority of failures — are solid.

Nothing through the product's sensor chain

CWRU and IMS are lab piezo recordings at constant speed and load. The product is a MEMS part on a magnet mount, duty-cycled, on a variable-load machine. That gap is unmeasured — closing it is the next milestone.

Closing this is cheap and mostly needs no hardware

IMS sets 1 and 3 are already downloaded (+3 failures). XJTU-SY, PRONOSTIA/FEMTO, Paderborn and MFPT take the run-to-failure count from 1 to ~35 and build a real healthy denominator. Weeks of work, zero capex.

Market — start in the Karaganda basin

Then Kazakhstan heavy industry, then CIS/EAEU

Anchor region. The Karaganda industrial basin is Kazakhstan's heavy-industry core: integrated steel, coal mines, iron-ore and coal-prep plants — thousands of monitored points of rotating equipment in non-explosive areas alone.

Expand. Kazakhmys, Kazzinc, ERG, cement plants, power stations, oil & gas processing.

Then CIS / EAEU — plants sharing the same equipment base, language, ERP (1C) and certification regime.

The budget is proven — locally

In March 2026 Qarmet (Temirtau) commissioned SPM Instrument's Intellinova Parallel EN online monitoring plus RDI Iris M, announced as an industry first for the region.

We read that as validation, not as a closed door. A tier-1 plant just proved that predictive-maintenance budget exists in this basin and that the buyer is sophisticated.

Our wedge is the tier *below* it: mid-market and legacy plants that will not sign a six-figure foreign platform contract, and the un-instrumented balance of plant at the large sites.

Competitive map, honestly drawn

The technique is public domain; the incumbents are real

TIER	WHO	WHERE THEY WIN	WHERE THEY LEAVE ROOM
Premium platform	Schaeffler OPTIME, SKF, Emerson, SPM, Augury	Tier-1 plants, full-service contracts, proven at scale	Price, contract size, English-first, remote support
Commodity sensor	Advantech (~\$340/node), Chinese vendors (\$33–189)	Unit price	No diagnosis — they ship numbers, not verdicts, and no local service
RU/CIS vendors	Auton, НПЦ Динамика, Вибро-Центр, ДИАМЕХ	Russian language, EAEU certification, some already KZ-registered	Not Kazakhstan-resident; no KZ local-content status; Kazakh-language support
Us	Bearing Health Monitor	Explainable per-element diagnosis, KZ residency, RU/KZ, on-site, mid-market pricing	No revenue, no field deployment, no certification — yet

The moat is not the algorithm

Envelope analysis is public-domain literature (Balderston 1969 → McFadden & Smith 1984 → Antoni 2007), and the patent that inspired this project expired in 2013. What compounds is **local presence, per-plant failure data, and the relationships** — not the DSP. We would rather state that than pretend otherwise.

Go to market: earn the right to sell hardware

Cash and evidence before capex and certification

1 · Paid diagnostics — now

Walk a plant with off-the-shelf instruments. Deliver a written report per asset naming the failing element.

Revenue in weeks. No imports, no certification, no customer capex.

Proves the two things that matter: will they pay, and can they act on a finding.

2 · Permanent nodes — 6–12 mo

Wire up the specific points where manual routes are impossible or too slow, at plants already paying for reports.

Off-the-shelf sensors first. Recurring revenue per monitored point. Failure dataset starts compounding.

3 · Own hardware — 12–24 mo

Locally assembled wireless node only once node volume justifies certification and tooling.

Unlocks KZ local-content preference with state-linked buyers. Ex certification opens underground mining.

Each stage is funded by the one before it, and each answers a question that would otherwise be discovered expensively at the next stage. The failure mode we are deliberately avoiding: becoming a small-scale importer-assembler before knowing whether a plant will pay.

What could kill this

And what we are doing about each

RISK	MITIGATION
The customer cannot act on the alert — no spare, no planner, no crew, so a warning creates no capturable value.	This is the first thing stage 1 tests, before any capex. If true, the business is diagnosis + parts + repair, not a sensor — and we would rather learn that for \$700 than for \$178k.
Supply chain — Digi-Key and Mouser will not ship to Kazakhstan; PJRC/SparkFun bar Teensy export.	Documented working routes (local importer, TME Poland, Chip One Stop). Stage 1 and 2 need no restricted parts at all; the exposure only begins at stage 3.
Solo founder across embedded, DSP, cloud, sales and certification.	Actively seeking a commercial co-founder with plant relationships. This is the constraint we rate highest.
False alarms destroy trust — one bad page can end a pilot.	Sustained-crossing logic (3 consecutive captures) already ships. The false-alarm denominator is the top engineering priority, and it is cheap to build.
Incumbents bundle monitoring free to sell bearings and lubricants.	We do not compete on unit price with a loss-leader. We compete on being here, in Russian and Kazakh, on legacy equipment the OEM does not care about.

Next 6 months

Sequenced so the cheapest business-killing question is answered first

WHEN	MILESTONE	QUESTION IT ANSWERS
Weeks 1–2	First paid diagnostic walkthrough at one Karaganda plant, 20–40 bearings, written report	Will anyone pay for the finding?
Weeks 1–4	Field rig assembled and validated against a seeded defect on a real machine	Does the MEMS chain see what the lab piezo chain saw?
Weeks 3–8	Validation expanded to ~35 run-to-failure events; healthy denominator built	What is the actual false-alarm rate and lead-time spread?
Month 3	Commercial co-founder onboarded; AIFC entity registered	Can we sell, and are we investable?
Months 3–6	Three paying plants on periodic diagnostics; first permanent nodes at one of them	Does this recur, and does it scale past one relationship?

What we want from an accelerator or first cheque

Not primarily money. **Introductions to plant maintenance directors**, help structuring a KZ/AIFC entity, and a commercial co-founder. The engineering risk on this project is the part that is already retired.

The physics works. The open question is the customer.

We have built and published the detection core, validated it on public data, and stated exactly where the evidence stops. The next milestone is one plant that pays — and that is what we are going after now.

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bearing-monitor.pages.dev · bearing-monitor.pages.dev/demo.html

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